AP PRECALCULUS

Unit 1 • Lesson 1.12 • Notes (filled in)

Topic 1.12 • Transformations of Functions

Notes (filled in)

Learning goals

- Describe translations, dilations and reflections, and their order.
- Write g in terms of f from a description, a table or a graph.
- Find the domain and range of a transformed function and the image of a point.

Vocabulary. translation, vertical dilation, horizontal dilation, reflection, image of a point, domain, range.

1. The four parameters

Key idea

For $g(x) = a\,f\big(b(x-h)\big) + k$:

- k: vertical translation, __up__ if $k > 0$. h: horizontal translation, __right__ if $h > 0$.
- a: vertical dilation by a factor of __$|a|$__; if $a < 0$, also a reflection over the __x-axis__.
- b: horizontal dilation by a factor of __$\frac{1}{|b|}$__; if $b < 0$, also a reflection over the __y-axis__.

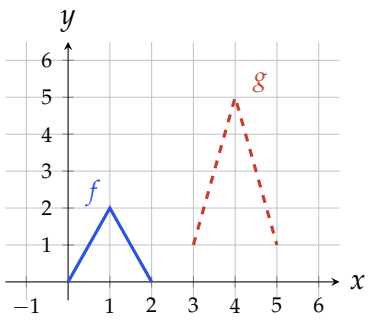
A point (x, y) on f moves to __$\left(\frac{x}{b} + h,\ ay + k\right)$__ on g. Horizontal changes affect the __domain__; vertical changes affect the __range__.

Example 1 • Describe the transformations

Describe how to obtain $g(x) = -2(x+3)^2 + 5$ from $f(x) = x^2$.

Shift left 3, vertical dilation by a factor of 2, reflection over the x-axis, shift up 5. The vertex moves from $(0, 0)$ to $(-3, 5)$.

Example 2 • From a graph



Write g in terms of f.

Same width; height above the base doubles ($2 \to 4$); moved right 3 and up 1: $g(x) = 2f(x-3) + 1$. Check: $2f(1) + 1 = 5$ at $x = 4$.

Example 3 • Domain, range and points

f has domain $[-4,2]$ and range $[-1,3]$, and $(2,3)$ is on its graph. For $g(x) = -f(2x)+1$, find the domain, the range and the image of $(2,3)$.

Domain: $2x \in [-4,2]$, so $[-2,1]$. Range: $-f \in [-3,1]$, then $+1$: $[-2,2]$.
Image of $(2,3)$: $x = \frac{2}{2} = 1$, $y = -3+1 = -2$: $(1,-2)$.

Example 4 • From a table

x	-2	-1	0	1	2
$f(x)$	5	2	1	2	5
$g(x)$	2	1	2	5	10

Write g in terms of f.

The values of g are those of f one input earlier: $g(-2) = f(-1)$, $g(-1) = f(0)$, $g(0) = f(1)$, $g(1) = f(2)$. So $g(x) = f(x+1)$: the graph of f shifted 1 unit left.

Common mistakes

- Shifting the wrong way: $f(x+3)$ moves the graph *left*.
- Thinking $f(2x)$ stretches horizontally; it compresses by a factor of $\frac{1}{2}$.

Check your understanding

1. $(2,5)$ is on f. Then $(1,2)$ is on $y = f(x+1)-3$.
2. $y = -\frac{1}{2}f(x)$ is a vertical dilation by $\frac{1}{2}$ and a reflection over the x-axis.